

HAPPIEST HEALTH DIGITAL TWIN CHALLENGE 2026

DoseTwin

Test the dose on your twin before you inject it.

A personal glucose–insulin digital twin for people with diabetes on insulin pens, validated on type 1. It predicts lows and personalises every meal dose from CGM, pen, step-counter and clinic data.

Team Drafts · Siva Sai Varshith Mitta, Abhijith Jayan · IIT Goa and KL University · Research prototype, evaluated in silico, not a medical device

Every insulin dose is a manual guess

9.4 lakh

people in India live with type 1 diabetes; about 3 lakh are under 20.

4–5×

insulin pen injections a day. Most families cannot afford pumps or automated delivery.

After

Affordable CGM sensors alarm only after glucose is already low, often at night.

Each dose depends on a carb guess, clinic ratios set months ago, and the day's activity. Too much insulin causes hypoglycaemia; too little causes long-term damage.

Why the bolus calculator gets it wrong

What the calculator assumes	What actually happens
Carbs are counted correctly	Counts are off by 20–50%, with a personal bias
Clinic ratios are right	Carb ratio and correction factor drift; reviews are months apart
Every day is the same	Walks, dawn rise and daily sensitivity change the need
Insulin on board is linear	Absorption differs by person, dose and timing

A twin that knows **this** person's physiology can answer the question the calculator cannot: **what will this dose do over the next five hours?**

DoseTwin does two jobs

ADVERSE EVENT

Predicts lows before they start

Warns about hypoglycaemia while there is still time to eat 15 g of carbs, including during sleep.

PERSONAL TREATMENT

Tests every dose before injection

Simulates the next 5 hours for each half-unit dose and recommends the lowest-risk dose that stays above a safety floor.

Inputs: CGM every 5 min · smart-pen doses · meal log · phone steps · synthetic EHR (diagnoses, labs, genetics, prescriptions).

Architecture

DEVICE DATA

CGM, every 5 min

Smart-pen doses

Meal log

Phone steps

Synthetic EHR: labs, Rx



PERSONAL TWIN

EHR prior: prescriptions and HbA1c set the starting physiology

MAP fit on the patient's own 5-hour forecasts, about 1.3 s

Observer locks the twin to every CGM reading: insulin and carbs on board, activity



Low alert

Twin 40-min forecast + CGM trend

Dose advisor

5-h what-if for every half unit; safety floor 70 mg/dL

Python · NumPy · Numba · SciPy · LightGBM baseline · browser twin in JavaScript (parity-tested to 1e-6 mg/dL)

AI and model details

Component	What it is
Twin model	10-state extended Bergman model: SC insulin, carb absorption, activity from steps, dawn effect
Personalisation	EHR prior (weight, insulin prescriptions, HbA1c), then MAP fit on 5–300 min forecast error
Synchronisation	Fixed-gain observer on CGM; unexplained-rate state captures surprises
Dose advisor	Kovatchev risk minimisation; rejects doses whose pessimistic path dips below 70
Low alert	Hybrid score: twin forecast minimum + 30-min CGM trend; alerts below 65
Baseline	LightGBM forecaster and classifier trained on 30 other patients

The twin is interpretable: every parameter has a physiological meaning a clinician can review, and it can answer **what if** questions that a pure ML forecaster cannot.

30 PATIENTS · HIGHEST RISK FIRST

Child, 10 y, M

child#005 · HbA1c 5.9%

<70 10.6%

TIR 77%

Child, 8 y, F

child#003 · HbA1c 6.2%

<70 10.1%

TIR 72%

Adult, 26 y, F

adult#006 · HbA1c 6.3%

<70 11.3%

TIR 69%

Teen, 19 y, M

adolescent#002 · HbA1c 7.8%

<70 6.9%

TIR 47%

Adult, 66 y, M

adult#004 · HbA1c 5.6%

<70 9.2%

TIR 78%

Teen, 17 y, F

adolescent#004 · HbA1c 6.6%

<70 6.5%

TIR 72%

Adult, 35 y, M

adult#007 · HbA1c 5.5%

<70 8.0%

TIR 89%

Child, 8 y, M

child#006 · HbA1c 6%

<70 7.8%

TIR 69%

Child, 9 y, M

child#002 · HbA1c 6.2%

<70 7.7%

TIR 81%

Child, 12 y, F

child#010 · HbA1c 6.7%

<70 6.3%

TIR 69%

Child, 12 y, F

child#004 · HbA1c 5.7%

<70 6.9%

TIR 87%

Child, 10 y, F

child#008 · HbA1c 8.4%

<70 4.4%

TIR 48%

Child, 10 y, male

child#005 · Kerala · T1D for 6 y · BMI 14.4

HbA1c 5.9%

77%

Time in range 70–180

On target · >70%

10.6%

Below 70

Above target · <4%

2.5%

Below 54

Above target · <1%

21.8%

Below 70, 00–07 h

Above target · <4%

40%

Variability (CV)

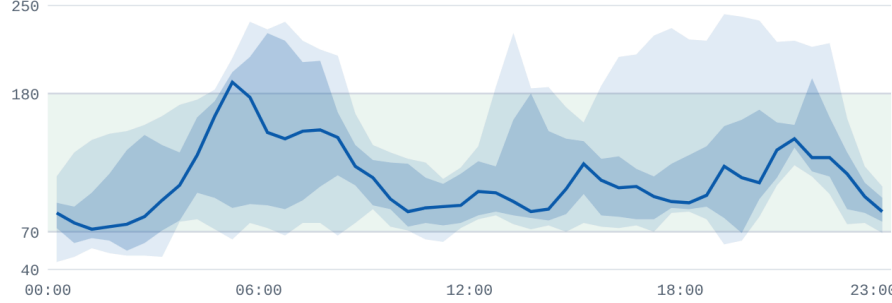
Above target · ≤36%

21

Low episodes, 7 days

Above target · fewer is better

GLUCOSE PROFILE, LAST 7 DAYS (MEDIAN, 25–75% AND 10–90% RANGES)



VIRTUAL PATIENT REPLAYS DAY 7 UNDER THE TRIAL SETTINGS



Twin prediction for day 7: time in range **93% → 96%** (+3), time below 70 **6.9% → 4.2%** (-2.8), lowest 61 → 61 mg/dL.
Same meals, carb counts and steps as the recorded day; doses recomputed with the trial settings. This is the twin's prediction, not a measurement.

DIAGNOSES

• Type 1 diabetes mellitus E10

LABS

HbA1c

5.9%

Fasting C-peptide

0.11 nmol/L

eGFR

93 mL/min

TSH

2.1 mIU/L

LDL

103 mg/dL

Urine ACR

8 mg/g

GENETICS

HLA-DR

DR4/DR4

GAD65 at diagnosis

positive

MEDICATIONS

• Insulin glargine: 14 U once daily

• Insulin aspart (pen): 1 U per 6 g carbohydrate; 1 U lowers glucose 30 mg/dL

Synthetic record. The twin uses weight, insulin prescriptions and HbA1c; the rest is context for the clinician.

SETTINGS REVIEW

Setting	Current	Twin estimate	Suggested
Carb ratio (g per U)	6.4	11.2	7.7 +20%
Correction (mg/dL per U)	30	33	33 +11%
Basal (U per day)	14.4	9.5	11.5 -20%

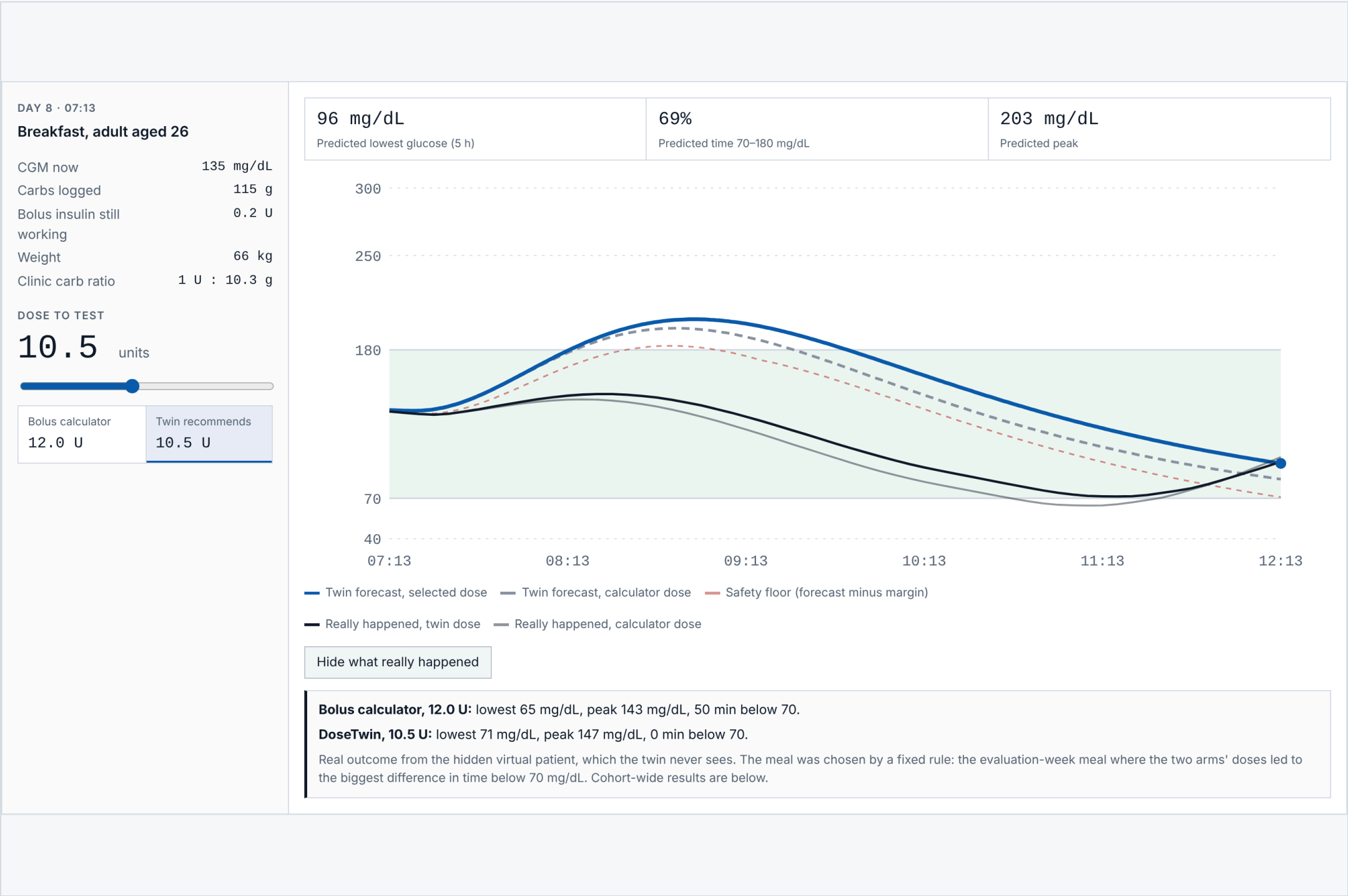
Suggestions move at most 20% per review, in the direction the twin indicates. When the twin and the record differ more than twofold, the twin is flagged as low confidence and the setting is kept.

The doctor's view

Patients ranked by hypoglycaemia risk, with the record, last week's glucose profile and the twin's suggested settings.

Move a setting and the virtual patient replays a real day under it.

Suggestions change at most 20% per review and are flagged when the twin is unsure.



Live demo

Drag the dose. The twin re-simulates the next 5 hours in the browser.

Here the calculator says 12 U; the twin says 10.5 U.

Real outcome: 50 min below 70 with 12 U, none with 10.5 U.

Example meal picked by a fixed, disclosed rule. Cohort results follow.

How we tested it without fooling ourselves

Hidden truth

30 virtual patients from the published UVA/Padova T1D model. The twin uses a different model and never sees it.

Realistic life

Late Indian dinners, 30% carb-count error, missed and late boluses, mis-set clinic ratios, walks, sensor noise.

Held-out cohorts

All settings frozen on a development cohort. Results come from 3 fresh cohorts: 90 patient-weeks.

Paired arms

Every arm relives the same 7 days. Only dosing and alerts differ. Bootstrap CIs and Wilcoxon tests.

Fewer lows, more time in range

−69%

time below 70 mg/dL (3.65% → 1.13%)

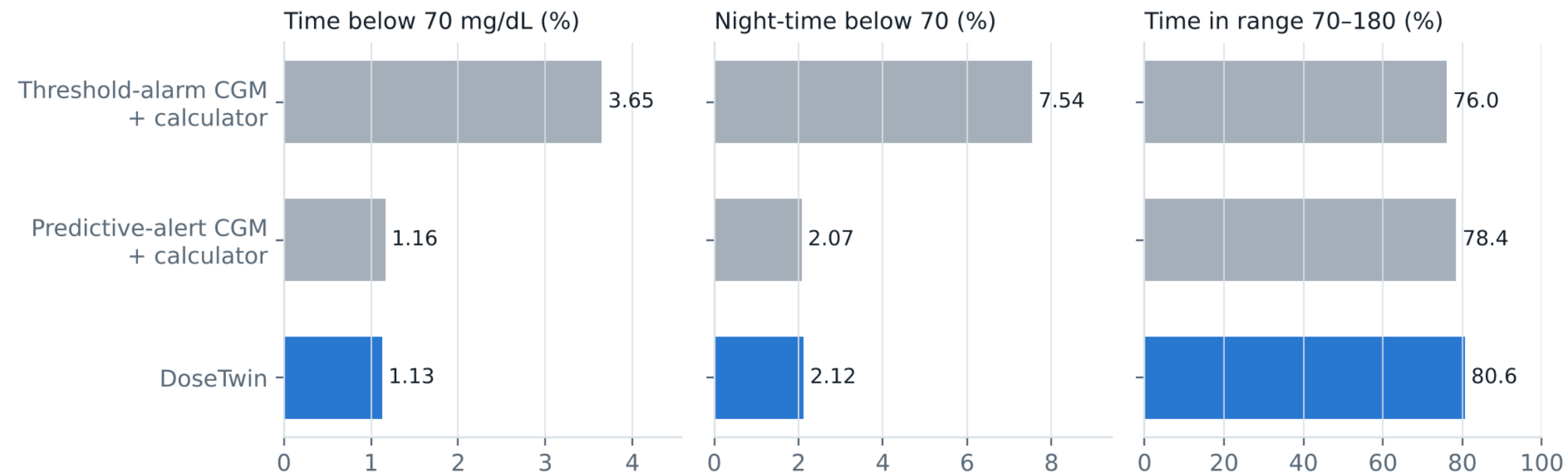
−72%

night-time below 70 (7.5% → 2.1%)

+4.6 pts

time in range (95% CI 2.9 to 6.4)

Outcomes on identical days, held-out cohorts (30 patients × 3)



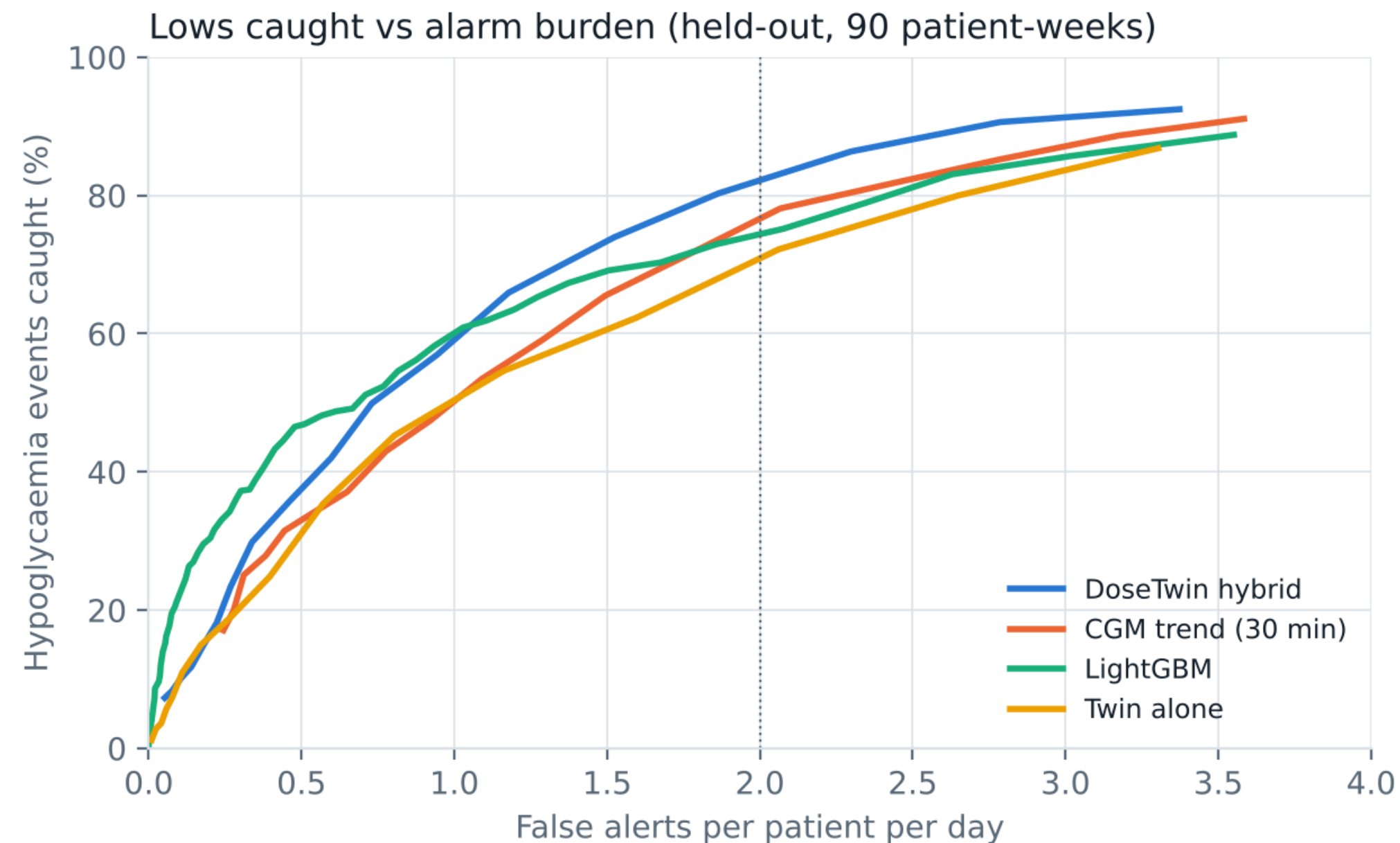
Vs threshold-alarm CGM + bolus calculator, all $p < 0.001$. Vs premium predictive-alert CGM: TIR +2.2 (CI 0.05 to 4.6, $p = 0.13$), lows no different.

Catching lows early

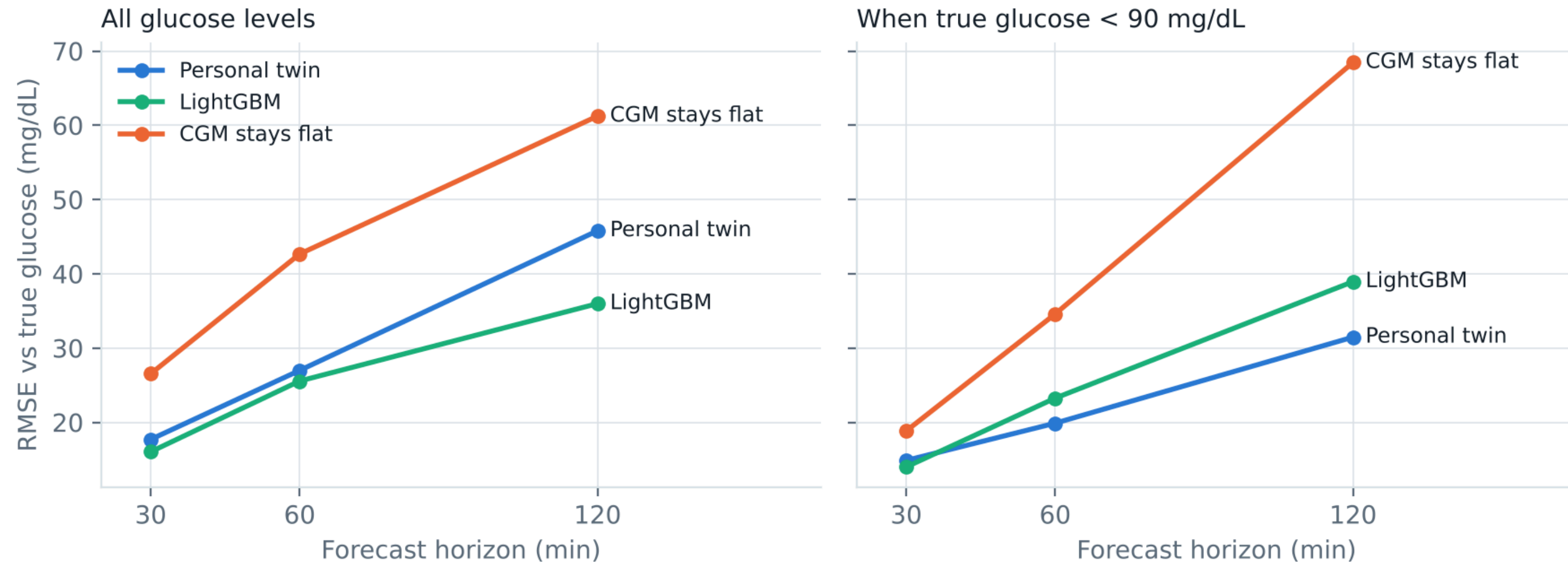
At ≤ 2 false alerts a day, DoseTwin caught **80%** of lows with a median 26 min warning.

CGM trend line: 71%. LightGBM trained on 30 patients: 73%.

LightGBM wins at ≤ 1 alert a day and warns earlier. The twin alone lost to the trend line; the hybrid beats both.



Most accurate where it matters: near lows



Below 90 mg/dL, 60-min error is 19.9 mg/dL for the twin vs 23.3 for LightGBM. Overall, LightGBM is as good or better, and better at 2 h. The twin learns from 7 days of one person's data and can simulate doses; LightGBM cannot.

What this does not prove yet

Limitation	What we found
Virtual patients only	Real-patient retrospective validation is the next step
Small dosing gain	+2.2 TIR vs premium CGM, not significant; perfect bolus adherence would add only ~2
Personalisation	With HbA1c in the prior, even an unpersonalised twin is fairly safe; personalising cuts alerts 4.2 → 2.6 a day
Settings recovery	Twin carb ratio closer to truth in 53% of cases; changes capped at 20% and clinician-approved
Easy cohort	Standard-care TIR ~76%, better than real life

Premium protection on an affordable sensor

For families

Earlier warnings at night and a second opinion on every dose, using the phone and sensor they already have.

For clinics

A twin-based review of carb ratio, correction factor and basal between visits, with every suggestion explained.

Roadmap

Real-data validation · CGM and smart-pen connectors · ABDM/FHIR export · paediatric clinic pilot

Built to be checked

Item	Detail
Stack	Python 3.11, NumPy, Numba, SciPy, pandas, LightGBM, Matplotlib; vanilla JavaScript demo
Tests	10 tests: simulator parity (< 0.5 mg/dL), twin, advisor safety, browser parity ($< 1e-6$)
Reproduce	One command, about 15 min on a laptop CPU; Dockerfile included
Speed	Twin fit about 1.3 s per patient; dose search is real-time in the browser
Licence	MIT; simulator parameters from simglucose (MIT)
Data	No real patient data used; nothing personal stored

Test the dose on your twin before you inject it.

69% less time in hypoglycaemia and 72% fewer night-time lows than an affordable threshold-alarm CGM, in a held-out in-silico study.